

SUBMISSION REQUIREMENT 1

Technical Approach

Proposed architecture, security model, workflow controls, and delivery stack for a South African multi-stage application system with POPIA compliance built in from day one.

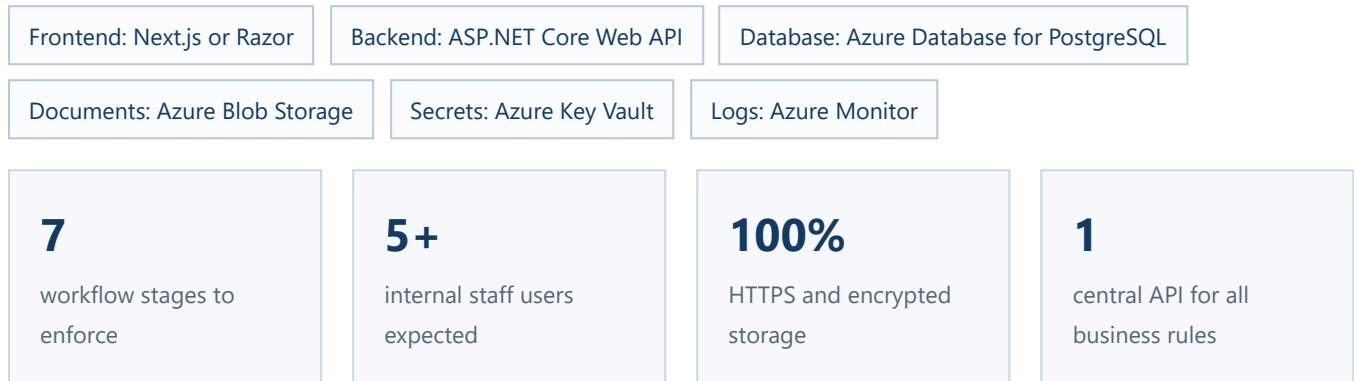
Project	Application Portal Demo and Proposal Pack
Recommended Hosting	Microsoft Azure, South Africa North (Johannesburg)
Document Date	4 May 2026

Recommendation summary: Azure South Africa North is preferred because it keeps primary application data at rest in South Africa while still supporting a standard enterprise .NET stack.

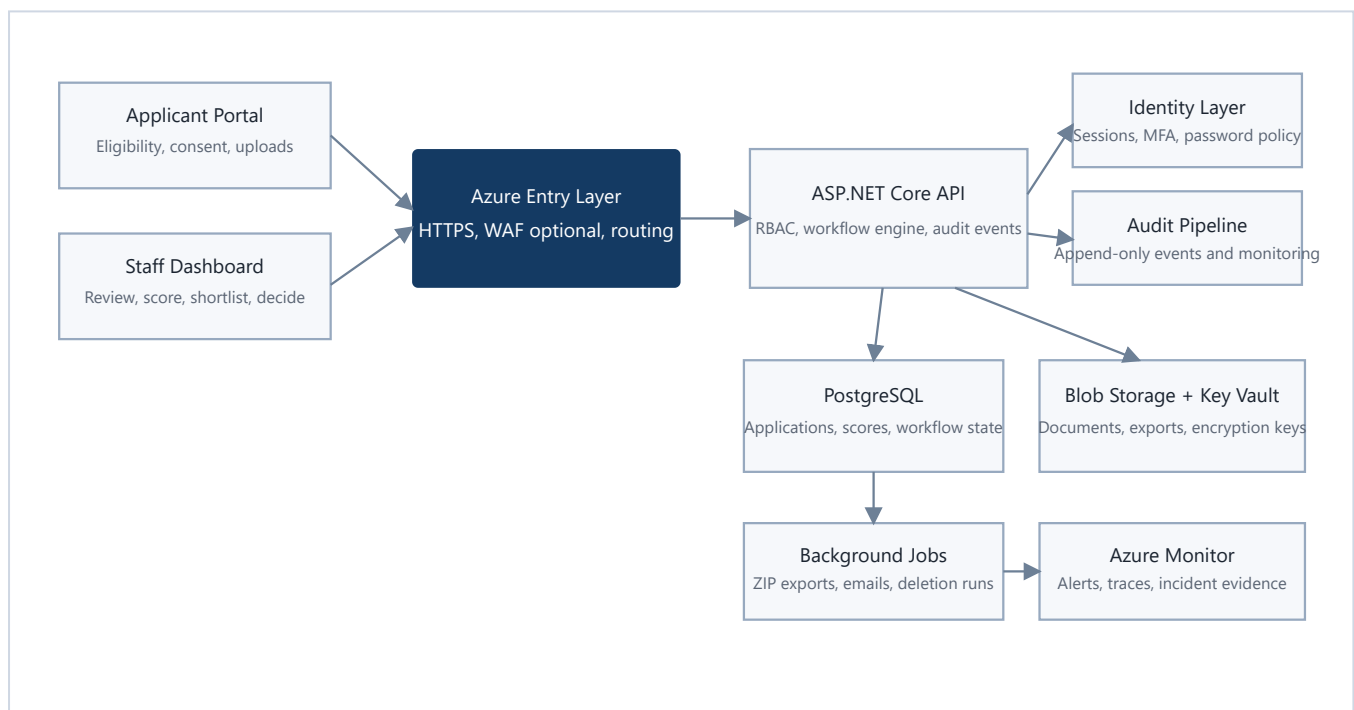
ARCHITECTURE

1. Proposed Platform

The recommended production stack is **.NET 8** for the backend, **PostgreSQL** for transactional workflow data, and **Azure South Africa North** for primary hosting. This combination fits the project because it handles strict role separation, transactional workflow gating, document-heavy intake, and detailed auditability without introducing unnecessary operational complexity.



2. High-Level Architecture



Design principle: all privileged actions flow through the API layer. Neither the browser nor direct database access is trusted to enforce permissions or workflow ordering.

SECURITY AND WORKFLOW

3. Role Access and Workflow Control

3.1 Role-Based Access Control

Access should be enforced in three layers: API authorization policies, query scoping, and database-level protections for the most sensitive reviewer tables. Roles should not simply open screens; they should grant specific capabilities.

Role	Visible Data	Explicit Restriction
Essay Reviewer	Essay answers, applicant summary, own score form	No academic marks, no peer reviewer scores, no shortlist view
Academic Reviewer	Transcript capture sheet, subject marks, average result	No peer essay scores unless separately authorized
Document Checker	Document checklist, file previews, issue notes	No editing of essay or academic scoring
Screener	Combined scorecard, ranking, document status, shortlist action	No rewrite of prior reviewer-owned records
Interview Admin	Shortlisted applicants, interview score sheet, notes	No access to earlier-stage edit controls
Final Decision Maker	Total score view, interview results, final outcome controls	No deletion of audit evidence

3.2 Workflow State Machine

State	Key Rule
EligibilityFailed	Application reference is retained for reporting, but full application cannot proceed.
Submitted	Mandatory documents and consent must already be present.
EssayReviewed	Essay review is stored with reviewer ID and timestamp.
AcademicsScored	Subject-level marks plus calculated average are preserved.
DocumentsVerified	Checklist completion and any issues are locked to the verification record.

Shortlisted	Blocked until essay, academics, and documents are complete.
InterviewScored	Allowed only for shortlisted applicants.
FinalApproved / FinalRejected / FinalStandby	Final outcome commits list generation and communication workflow.

3.3 Concurrency and Duplicate Prevention

Submission Safety

- Unique active application per applicant per cycle.
- Idempotency token on the submit endpoint.
- Mandatory document validation before the status can become Submitted.

Reviewer Safety

- Optimistic concurrency token on application stage records.
- Unique review rows per reviewer and stage.
- Task assignment locks for shortlist and final decision actions.

Practical effect: if two staff users try to save the same stage decision at the same time, the second save is rejected and forced to refresh before proceeding.

POPIA CONTROLS

4. POPIA Implementation Design

4.1 Core Compliance Controls

Consent and Purpose

- Explicit consent recorded before personal data submission.
- Consent version, timestamp, IP address, and policy text hash retained.
- Purpose wording tied to the application cycle and programme rules.

Encryption

- TLS enforced on all browser and API traffic.
- PostgreSQL encrypted at rest.
- Blob storage encrypted at rest, with secrets and keys in Key Vault.

Audit Logging

- Every sensitive read/write action generates an audit event.
- Actor, role, target entity, timestamp, and change summary are captured.
- High-risk downloads and exports are logged separately.

Data Subject Rights

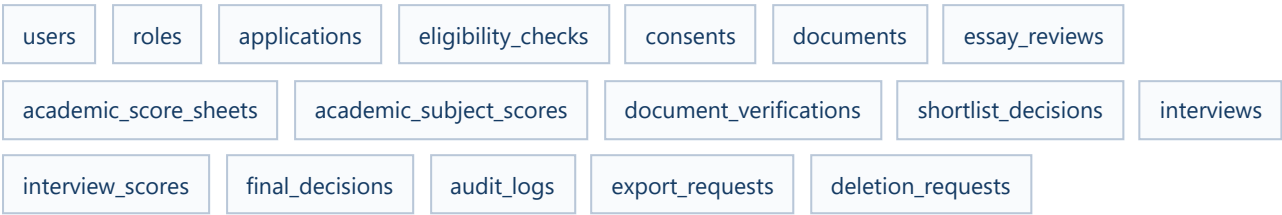
- Self-service or admin-assisted export package generation.
- Correction, deletion, and post-cycle retention workflows.
- Breach response playbook linked to regulator notification procedures.

4.2 Audit Logging Model

Event Type	Examples	Required Fields
Authentication	Login, logout, failed login, password reset	User ID, timestamp, IP, user agent, result
Data Access	Document download, export request, applicant record view	User ID, target record, purpose/action, correlation ID
Workflow Changes	Essay score saved, shortlist decision, final approval	Before/after summary, acting role, affected application
Security Events	Role change, permission change, suspicious access alert	Actor, target user, reason, timestamp, environment

4.3 Database Structure

The data model should separate each stage into distinct tables so the system can enforce ownership, preserve audit history, and apply narrower permissions.



Database-level enforcement should be used selectively for the most sensitive reviewer data. The main control plane remains the application service layer, but PostgreSQL row-level security can be added for reviewer-owned rows and privileged read paths.